

Assignment 22.3 Ai Assisted Coding

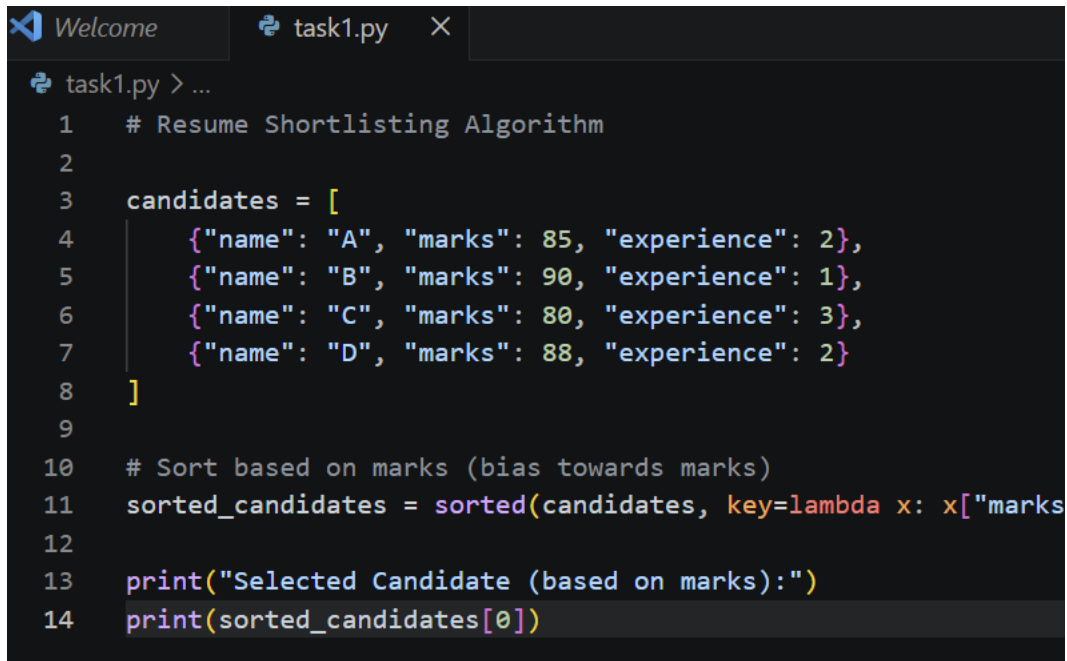
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Task 1: Bias Detection in AI Suggestions

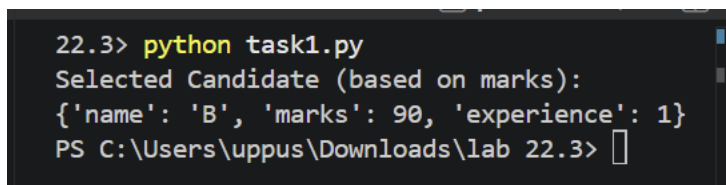
Prompt: Generate a resume shortlisting algorithm to select the best candidate.

Code:



```
task1.py > ...
1  # Resume Shortlisting Algorithm
2
3  candidates = [
4      {"name": "A", "marks": 85, "experience": 2},
5      {"name": "B", "marks": 90, "experience": 1},
6      {"name": "C", "marks": 80, "experience": 3},
7      {"name": "D", "marks": 88, "experience": 2}
8  ]
9
10 # Sort based on marks (bias towards marks)
11 sorted_candidates = sorted(candidates, key=lambda x: x["marks"])
12
13 print("Selected Candidate (based on marks):")
14 print(sorted_candidates[0])
```

Output:



```
22.3> python task1.py
Selected Candidate (based on marks):
{'name': 'B', 'marks': 90, 'experience': 1}
PS C:\Users\uppus\Downloads\lab 22.3> 
```

Explanation:

- Algorithm selects candidate based only on marks.
- It ignores other factors like diversity or experience.
- This shows how AI can be biased.

Task 2: Privacy and Sensitive Data Handling

Prompt: Write a program to process student records including roll number, marks, and email. Ensure data privacy.

Code:

```
1  # Student Record Processing with Data Privacy
2
3  students = [
4      {"roll": 101, "marks": 85, "email": "student1@gmail.com"},
5      {"roll": 102, "marks": 90, "email": "student2@gmail.com"}
6  ]
7
8  # Function to mask email
9  def mask_email(email):
10     parts = email.split("@")
11     return parts[0][0] + "***@" + parts[1]
12
13  print("Student Records (Protected Data):")
14
15  for s in students:
16     print("Roll:", s["roll"])
17     print("Marks:", s["marks"])
18     print("Email:", mask_email(s["email"]))
19     print("-----")
```

Output:

```
PS C:\Users\uppus\Downloads\lab 22.3> python task2.py
Student Records (Protected Data):
Roll: 101
Marks: 85
Email: s***@gmail.com
-----
Roll: 102
Marks: 90
Email: s***@gmail.com
-----
PS C:\Users\uppus\Downloads\lab 22.3>
```

Explanation:

- Sensitive data like email is masked for privacy.
- Personal information should not be fully visible.
- Developers must protect user data using secure methods.

Task 3: AI-Generated Security Risks

Prompt: Generate a payment processing system code for an online shopping website.

Code:

```
task3.py > ...
1  # Secure Payment Processing Example
2
3  def validate_card(card_number):
4      # Check if card number is 16 digits
5      return card_number.isdigit() and len(card_number) == 16
6
7  def process_payment(card_number, amount):
8      if not validate_card(card_number):
9          print("Invalid Card Details!")
10         return
11
12         # Mask card number (security)
13         masked_card = "**** *  " + card_number[-4:]
14
15         print("Processing payment...")
16         print("Card Used:", masked_card)
17         print("Amount Paid:", amount)
18         print("Payment Successful!")
19
20     # User Input
21     card = input("Enter 16-digit card number: ")
22     amt = float(input("Enter amount: "))
23
24     process_payment(card, amt)
```

Output:

```
PS C:\Users\uppus\Downloads\lab 22.3> python task3.py
Enter 16-digit card number: 1234567812345678
Enter amount: 12000
Processing payment...
Card Used: **** *  5678
Amount Paid: 12000.0
Payment Successful!
PS C:\Users\uppus\Downloads\lab 22.3>
```

Explanation:

- Card details are validated before processing.
- Full card number is hidden (masked) for security.
- Prevents risks like data theft and fraud.

Task 4: Accountability in Human–AI Collaboration

Prompt: Generate code for an autonomous traffic signal control system using AI. Also explain who is responsible if the system causes an accident.

Code:

```
task4.py > ...
1  # Autonomous Traffic Signal Control System (Simulation)
2
3  def traffic_signal_control(vehicle_count):
4      if vehicle_count > 50:
5          return "Green Light (Heavy Traffic)"
6      elif vehicle_count > 20:
7          return "Yellow Light (Moderate Traffic)"
8      else:
9          return "Red Light (Low Traffic)"
10
11 # Simulated input
12 vehicles = int(input("Enter number of vehicles: "))
13
14 result = traffic_signal_control(vehicles)
15 print("Signal Status:", result)
```

Output:

```
PS C:\Users\uppus\Downloads\lab 22.3> python task4.py
task4.py
Enter number of vehicles: 30
Signal Status: Yellow Light (Moderate Traffic)
PS C:\Users\uppus\Downloads\lab 22.3> 
```

Explanation:

- AI suggests logic but humans must verify and test the system.
- Software engineers are responsible for errors in the code.
- Government or organization deploying the system is finally accountable.

Task 5: Limitations of AI in Critical Systems

Prompt: Generate code for an aircraft maintenance fault detection system.

Code:

```

task5.py > ...
1  # Aircraft Fault Detection System (Simulation)
2
3  def detect_fault(temperature, pressure):
4      if temperature > 100 or pressure > 200:
5          return "Fault Detected! Maintenance Required"
6      else:
7          return "System Normal"
8
9  # User Input
10 temp = float(input("Enter engine temperature: "))
11 press = float(input("Enter pressure level: "))
12
13 result = detect_fault(temp, press)
14 print("Status:", result)

```

Output:

```

TERMINAL powershell
PS C:\Users\uppus\Downloads\lab 22.3> python task5.py
Enter engine temperature: 120
Enter pressure level: 180
Status: Fault Detected! Maintenance Required
PS C:\Users\uppus\Downloads\lab 22.3>

```

Explanation:

- AI-generated code may not handle all real-world situations.
- Human experts must validate and test the system.
- Safety-critical systems require strict checking and certification.

Task 6: Ethical Use of AI-Generated Code

Prompt: Explain the ethical issues of using AI-generated code in academic projects.

Code:

```
1 # Ethical Example: Student Marks Calculation
2
3 def calculate_average(marks):
4     return sum(marks) / len(marks)
5
6 # Input
7 marks = [80, 85, 90]
8
9 avg = calculate_average(marks)
10 print("Average Marks:", avg)
```

Output:

```
✓ TERMINAL
PS C:\Users\uppus\Downloads\lab 22.3> python task6.py
Average Marks: 85.0
PS C:\Users\uppus\Downloads\lab 22.3> 
```

Explanation:

- Using AI code without understanding is unethical.
- It may lead to plagiarism or copyright issues.
- Proper citation and manual verification are required.